

Croup - Summary

Key Symptoms: Barking cough (key indicator), fever, hoarse voice, laboured/noisy breathing, etc.

Risk Factors: age 6 months-3 years, male

Most common causing-pathogens: Viral cause: parainfluenza type 1&3, also others such as influenza A & B, adenovirus, respiratory syncytial virus (RSV), metapneumovirus, coronavirus and mycoplasma

Non-Pharmacological Treatment: Keeping the child calm, breathing cool air, oxygen (inpatient setting)

Pharmacological Treatment:

- Corticosteroids – mainstay
 - Dexamethasone PO is preferred
 - Nebulized budesonide is as effective, but use is limited due to cost
- Nebulized epinephrine – in severe cases
- Analgesics – for pain and fever

Monitoring Parameters:

- Onset of effect: 2-3 hours after corticosteroid therapy
- Review slides 17-18 to see when additional doses and/or hospitalization is required in moderate and severe cases. ~2 hours with no improvement in severe cases for additional doses, ~4 hours with no improvement in moderate (corticosteroid) and severe (corticosteroid & nebulized epinephrine) cases for hospitalization. Consider moving to the intensive care unit (ICU) if recurrent episodes of agitation or lethargy.

Additional Key Points:

- Inhaled Beta-2 Agonists (e.g. salbutamol), antibiotics, antitussives and decongestants are not effective in the treatment of croup and should NOT be used
- Avoid use of corticosteroids in children with a known immunodeficiency or recent exposure to varicella